



Research paper

Pharmacokinetic comparison of seven major bio-active components in normal and blood deficiency rats after oral administration of Danggui Buxue decoction by UPLC-TQ/MS



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ABSTRACT

Ethnopharmacological relevance: Blood deficiency is commonly encountered among women, and is the root of many gynecological disorders. Danggui Buxue Decoction (DBD), a classical traditional Chinese formula which is composed of Astragali Radix (AR) and Angelicae Sinensis Radix (ASR) at the ratio of 5:1 (w/w), is widely used in TCM clinics for treatment of blood deficiency syndrome. This study is to compare the *in vivo* pharmacokinetic properties of seven major bio-active components in normal and blood deficiency rats after oral administration of DBD.

Materials and methods: Blood deficiency rats were induced by bleeding from orbit at the dosages of 5.0 mL/kg each day for 12 days. Normal and blood deficiency rats were administered of DBD on the 12th day at the dosage of 20 g/kg, and blood was collected at different time points after then. Concentrations of ferulic acid, caffeic acid, butylphthalide, ligustilide, calycosin-7-O- β -glucoside, ononin, and astragaloside IV in plasma were quantified by UPLC-TQ/MS, and the main pharmacokinetic parameters were calculated by DAS 2.0.

Results: It was found that C_{max} , T_{max} and $MRT_{0\sim T}$ of astragaloside IV, C_{max} , $T_{1/2Z}$, $AUC_{0\sim T}$ and $MRT_{0\sim T}$ of calycosin-7-O- β -glucoside, $T_{1/2Z}$ and $AUC_{0\sim T}$ of ferulic acid, $T_{1/2Z}$, $AUC_{0\sim T}$ and $MRT_{0\sim T}$ of ononin, and $MRT_{0\sim T}$ of ligustilide, butylphthalide, and caffeic acid in blood deficiency rats was significantly different ($P < 0.05$) from normal rats.

Conclusions: This study was the first report about pharmacokinetic investigation in blood deficiency animals which was conducted by bleeding. And the results demonstrated that the seven DBD constituents in normal and blood deficiency rats had obvious differences in some pharmacokinetic characteristics, suggesting that the rate and extent of drug metabolism were altered in blood deficiency animals.

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1. Introduction

Based on theories of traditional Chinese medicine (TCM), blood deficiency was a common syndrome characterized by pallor complexion, palpitation or feeble pulse, and was considered to be the root of many gynecological disorders. With the increasing pace of life, many men began to suffer from it. This syndrome was often induced by massive blood loss, deficient spleen and stomach, insufficient hematogenesis and blood stasis. According to deficiency theory of TCM and earlier reports, blood loss was similar to blood deficiency of TCM (Shi et al., 2013), such as postoperative

and postpartum women with chronic bleeding, excessive menstruation, long menstrual period, or uterine bleeding. Among hundreds and thousands of TCM formulae, Danggui Buxue Decoction (DBD) is one representative formula for the treatment and prevention of blood deficiency in clinical medication. Recorded in “*Nei Wai Shang Bian Huo Lun*” (Jin Dynasty, AD 1247), DBD consisted of Astragali Radix (AR) and Angelicae Sinensis Radix (ASR) in the ratio of 5:1 on a dry weight basis and was recommended as tonic reinforcing “Qi” (vital energy) and blood in TCM clinical practice (Wang and Liang, 2010). Many pharmacological studies also showed that DBD could nourish and tonify blood (Shi et al., 2012; Yang et al., 2009; Gao et al., 2008; Zheng et al., 2010), and inhibit hepatic fibrosis, increase insulin sensitivity, activate extracellular signal-regulated kinase in cultured T-lymphocytes, and had other effects such as anti-oxidation and estrogenic

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properties (Wang and Liang, 2010; Gao et al., 2007a; Dong et al., 2006; Chiu et al., 2007; Gao et al., 2007b; Choi et al., 2011; Liu et al., 2011; Meng et al., 2010; Zheng et al., 2012).

Pharmacological studies and chemical investigation of the medicinal materials had found that butylphthalides, organic acids and their esters, and polysaccharides were the major beneficial components in ASR (Yi et al., 2009), while flavones and saponins in AR (Wang et al., 2006; Yu et al., 2007). By comparing the biological effects of DBD in stimulating osteoblast proliferation, estrogen promoter activation, and anti-platelet aggregation activity, the drug ratio of 5:1 represented the best effects. Interestingly, higher amounts of AR-derived astragaloside IV, calycosin and formononetin, and ASR-derived ferulic acid were found in DBT with AR and ASR in this ratio (Dong et al., 2006). Bioactivities of these marker compounds were also well described in earlier publications (Zhang et al., 2008; Wen et al., 2007), hence some pharmacokinetic strategies and platforms were established for better understanding of their *in vivo* course or metabolic disposition (Wen et al., 2008; Zhang et al., 2006; Zhao et al., 2003; Niu et al., 2008; Konishi et al., 2005; Zhang et al., 2011). However, there were fewer reports about their pharmacokinetic profiles in model animals whose pharmacokinetic process was usually different from that in normal animals (Taguchi et al., 2013).

Many researches demonstrated that disease condition would alter the pharmacokinetic parameters (Tian et al., 2010). The aim of present study was to compare pharmacokinetic process of seven major bioactive components including ferulic acid, caffeic acid, butylphthalide, ligustilide, calycosin-7-O- β -glucoside, ononin and astragaloside IV in normal and blood deficiency rats after oral administration of DBD by ultra-performance liquid chromatography coupled with a triple quadrupole electrospray tandem mass spectrometry (UPLC-TQ/MS). Moreover, the variation mechanism of nourishing and tonifying blood after oral administration of DBD was also possibly revealed.

2. Materials and methods

2.1. Chemicals and reagents

The radix of *Angelica sinensis* (Oliv.) Diels (Apiaceae) was collected at Min County, Gansu Province, China, in September 2011. The radix of *Astragalus membranaceus* (Fisch.) Bge. var. *mongolicus* (Bgr.) Hsiao (Fabaceae) was collected at Inner Mongolia, China, in September, 2011. They were identified by Dr. Hui Yan (Department of Pharmacognosy, Nanjing University of Chinese Medicine, Nanjing, China) and were up to the standard of Chinese Pharmacopoeia. The voucher numbers (No. NJUTCM-20110923 for Apiaceae, No. NJUTCM-20110924 for Apiaceae) were deposited in the Herbarium of Nanjing University of Chinese Medicine. Reference standards of ferulic acid (**1**, 98%), caffeic acid (**2**, 98%), butylphthalide (**3**, 98%), ligustilide (**4**, 98%), calycosin-7-O- β -glucoside (**5**, 98%), ononin (**6**, 98%), astragaloside IV (**7**, 98%), chloroamphenicol (**IS**₁) and clarithromycin (**IS**₂) were purchased from the National Institute for the Control of Pharmaceutical and Biological products (Beijing, China). Their chemical structures were presented in Fig. 1. Acetonitrile and methanol of HPLC grade were purchased from Tedia (Fairfield, OH, USA). Formic acid of HPLC grade was purchased from Merck (Darmstadt, Germany). Deionized water was distilled and purified by an EPED super-purification system (Eped, Nanjing, China).

2.2. Apparatus

ACQUITY™ UPLC system, Xevo™ TQ mass spectrometry system (Waters Corporation, Milford, USA); EPED ultra pure water machine (Nanjing, China); Sartorius BT1250 electronic balance (Sartorius Scientific Instruments Corporation, Beijing, China); TDL240B centrifuge (Anting Scientific Instrument Corporation, Shanghai, China),

WH-1 micro-vortex mixing instrument (Shanghai, China); CENTRI-VAP centrifuge enrichment apparatus (Labconco).

2.3. UPLC-MS/MS conditions

Chromatographic separation was carried out on an Acquity UPLC BEH C₁₈ column (2.1 × 100 mm², 1.7 μ m, Waters Corporation, Milford, USA) maintained at 35 °C. The mobile phase gradient conditions consisted of 0.1% aqueous formic acid (A) and acetonitrile (B): 0–7 min, 10–70% B; 7–8 min, 70% B; at 8–9 min, 70–10% B; 9–10 min, 10% B. The flow rate was 0.4 mL/min. The autosampler was conditioned at 4 °C and the sample injection volume was 4 μ L. The ESI source was set in both positive and negative ionization mode. The scanning mode was set multiple reaction monitoring (MRM) mode and the selected monitor ion were m/z 193 → m/z 134 for ferulic acid, m/z 1814 m/z 89 for caffeic acid, m/z 189 → m/z 128 for butylphthalide, m/z 1908 m/z 91 for ligustilide, m/z 447 m/z 285 for calycosin-7-O- β -glucoside, m/z 431 m/z 269 for ononin, m/z 784 → m/z 89 for astragaloside IV, m/z 749 → m/z 158 for clarithromycin, m/z 321 → m/z 152 for chloramphenicol. The TQ mass spectrometer was operated with a capillary voltage of 3 kV, a sampling cone voltage of 30 V, a source temperature of 150 °C, and a desolvation temperature of 550 °C. The MassLynx and DAS 2.0 software were employed to analyze the data.

2.4. Preparation of DBD extract

Two crude herbs including AR (300 g) and ASR (60 g) were thoroughly decocted in water for 2 h (1:8, w/v), and then the gruffs decocted twice with water (1:6, w/v), 1.5 h for each time, filtered through gauze. The three filtrates were merged and evaporated by rotary evaporation under vacuum at 60 °C and concentrated to 1.0 g/mL.

2.5. Preparation of calibration standards and quality control (QC) samples

The mixture of standard stock solution containing above seven components were prepared in methanol and given a final concentrations of 354 μ g/mL for ferulic acid, 38 μ g/mL for caffeic acid, 262 μ g/mL for butylphthalide, 730 μ g/mL for ligustilide, 291 μ g/mL for calycosin-7-O- β -glucoside, 338 μ g/mL for ononin, and 319 μ g/mL for astragaloside IV, respectively. The mixture stock solution was serially diluted with methanol to provide working standard solutions of desired concentration of 44.25 μ g/mL for ferulic acid, 47.86 μ g/mL for caffeic acid, 32.75 μ g/mL for butylphthalide, 91.25 μ g/mL for ligustilide, 36.38 μ g/mL for calycosin-7-O- β -glucoside, 42.25 μ g/mL for ononin, and 39.86 μ g/mL for astragaloside IV. The mixture of standard working solution was diluted into nine different concentration gradients. The **IS**₁ and **IS**₂ stock solution of 1000 μ g/mL for chloroamphenicol and 1000 μ g/mL for clarithromycin were prepared in methanol. The mixture of **IS**₁ and **IS**₂ working solution were diluted at the concentration of 3.3 ng/mL for clarithromycin and 33 ng/mL chloroamphenicol into methanol, which was called **IS**-Methanol solution (**IMS**).

Calibration standards and quality control (QC) samples were prepared as following: the mixture of standard working solution was diluted into nine different concentration gradients. And their given final concentrations were 0.354–354 ng/mL for ferulic acid, 0.385–385 ng/mL for caffeic acid, 0.262–262 ng/mL for butylphthalide, 0.73–730 ng/mL for ligustilide, 0.291–291 ng/mL for calycosin-7-O- β -glucoside, 0.338–338 ng/mL for ononin, 0.319–319 ng/mL for astragaloside IV. The QC samples were prepared at concentrations of 17.7, 35.4 and 177 ng/mL for ferulic acid, 19.25, 38.5 and 192.5 ng/mL for caffeic acid, 13.1, 26.2 and 131 ng/mL for butylphthalide, 36.5, 73 and 365 ng/mL for ligustilide, 14.55, 29.1 and 145.5 ng/mL for

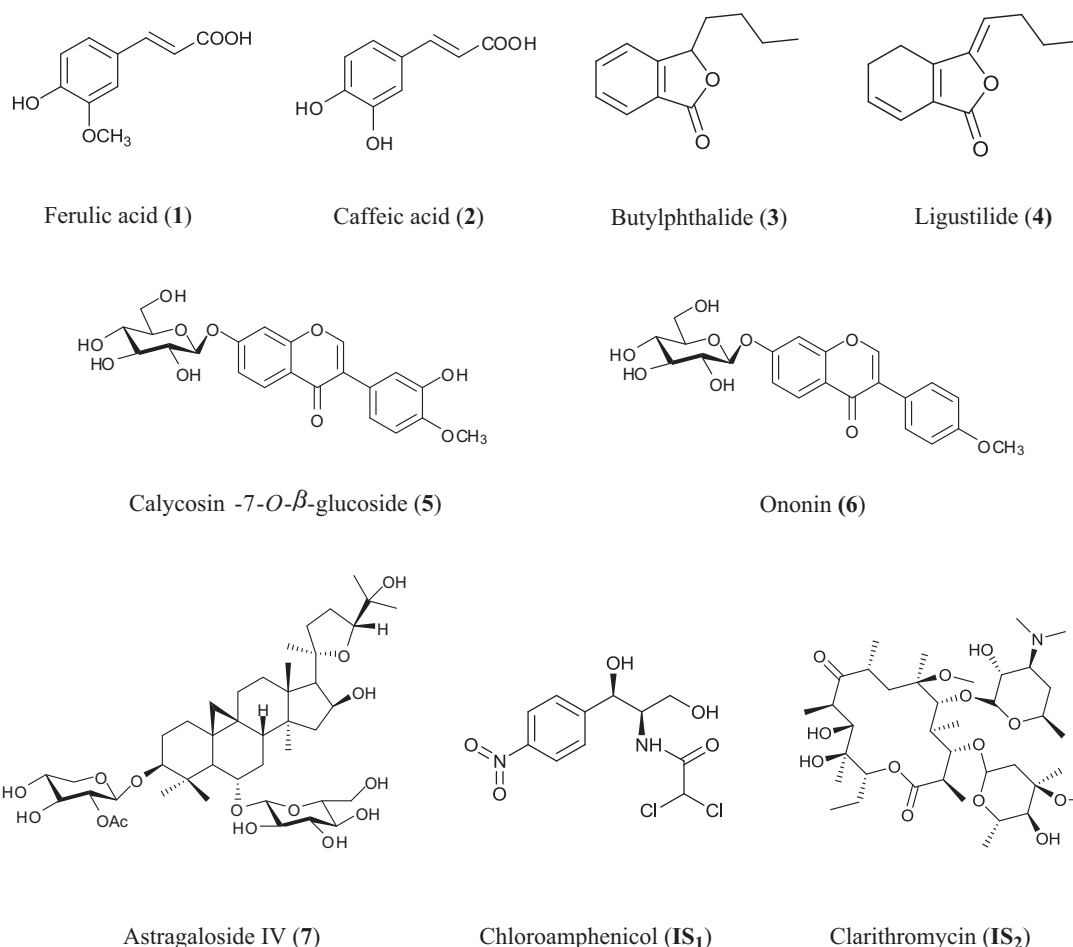


Fig. 1. Chemical structures of ferulic acid (1), caffeic acid (2), butylphthalide (3), ligustilide (4), calycosin-7-O-β-glucoside (5), ononin (6), astragaloside IV (7), chloroamphenicol (IS₁) and clarithromycin (IS₂).

calycosin-7-O-β-glucoside, 16.9, 33.8 and 169 ng/mL for ononin, 15.95, 31.9 and 159.5 ng/mL for astragaloside IV. All solutions were stored at 4 °C and brought to room temperature before use. QC samples were stored at –20 °C.

2.6. Pharmacokinetic study

Male and female Sprague-Dawley (SD) rats (180–220 g) were obtained from Zhejiang Province laboratory Animal Center and kept in an environmentally controlled breeding room (temperature: 20 ± 2 °C, humidity: 60 ± 5%) for two weeks before the experiments started. Animal welfare and experimental procedures were strictly in accordance with the *Guide for the Care and Use of Laboratory Animals* (US National Research Council, 1996) and the related ethics of Nanjing University of Chinese Medicine.

The blood deficiency rats were induced by bleeding at the account of 5.0 mL/kg body weight every day, and the experimental period was 12 days. The bleeding was performed by collecting blood from orbit (Shi et al., 2013).

24 SD rats were randomly divided into normal group and model group. The rats were fasted for 12 h with free access to water prior to the experiment. At the final day of experimental period, both normal and blood deficiency rats were orally administrated by DBD at a dosage of 20 g/kg body weight, and then 0.5 mL blood samples were collected into the centrifuge tubes which contained 0.05 mL EDTA-2Na at different time points (0, 5, 10, 15, 30, 45, 60, 120, 240, 360, 600, 720, and 1440 min). Blood samples were centrifuged at

13,000 rpm for 10 min at 4 °C and the supernatant was transferred and stored at –70 °C until analysis.

2.7. Sample preparation

600 μL IMS was added to 200 μL of plasma samples in a 1.5 mL centrifuge tube. The mixture was vortexed for 5 min and centrifuged at 13,000 rpm for 10 min at 4 °C. The supernatant was transferred into a new 1.5 mL centrifuge tube and concentrated in the centrifuge enrichment apparatus. The residue was reconstituted in 200 μL methanol and centrifuged as described above, and then 4 μL of the supernatant was injected into the UPLC-TQ/MS system for analysis.

2.8. Method validation

2.8.1. Specificity

The specificity was calculated by comparing blank plasma samples from normal rats and blood deficiency rats, blank plasma samples spiked with reference standards and internal standards, and plasma samples after oral administration of DBD. Blank plasma samples were analyzed for endogenous interference, followed by spiking with IS for the interference of IS.

2.8.2. Linearity and lower limit of quantification (LLOQ)

The linearity of each calibration curve was determined by plotting the peak area ratio (Y) of analytes to corresponding IS versus the nominal concentration (X) of analytes with weighted

($1/X^2$) least square linear regression. The lower limit of quantification (LLOQ) was determined as the lowest concentration with a signal-to-noise (S/N) ratio of 10.

2.8.3. Precision and accuracy

The precision and accuracy of the method were investigated by performing replicate analyses of QC samples against calibration standards. The precision was determined from inter-day and intra-day using five sets of low, middle, and high concentrations. The precision was expressed by relative standard deviation (RSD%), and the accuracy by relative percentage error (RR%).

2.8.4. Extraction recovery and matrix effect

The extraction recovery was determined by calculating the ratio of the amounts of QC samples finally obtained against those originally spiked in drug-free plasma. The matrix effect was calculated by comparing the peak areas of post-extraction plasma spiked with analytes to those prepared in re-dissolve solution.

2.8.5. Stability

The stability of the method was determined from different conditions using five sets of low, middle, and high concentration. The short-term stability was tested after placing the samples at room temperature for 12 h and at refrigerated (4 °C) for 24 h. The freeze-thaw stability was tested by freezing the samples at –70 °C overnight, and thawing at 20 °C, for a total of three cycles. The long-term stability was tested after keeping the samples at low temperature (–70 °C) for 14 days.

2.9. Statistics

The concentration–time data were analyzed by Masslynx 4.1 (provided by Waters Corporation, Milford, USA) to obtain the pharmacokinetic parameters of ferulic acid, caffeic acid, butylphthalide, ligustilide, calycosin-7-*O*- β -glucoside, ononin and astragaloside IV, and statistical analysis was SPSS 16.0. An independent-samples *T*-test was used for the comparison. A value of $P \leq 0.05$ was considered significantly difference.

3. Results and discussion

3.1. Specificity

No interference was observed at the retention times of either analytes or IS in any plasma samples which is used for analysis, and the method exhibited good specificity. The typical MRM chromatograms were shown in Fig. 2.

3.2. Linearity and LLOQ

The linear regression equation, correlation coefficient and LLOQ for ferulic acid, caffeic acid, butylphthalide, ligustilide, calycosin-7-*O*- β -glucoside, ononin and astragaloside IV were shown in Table 1.

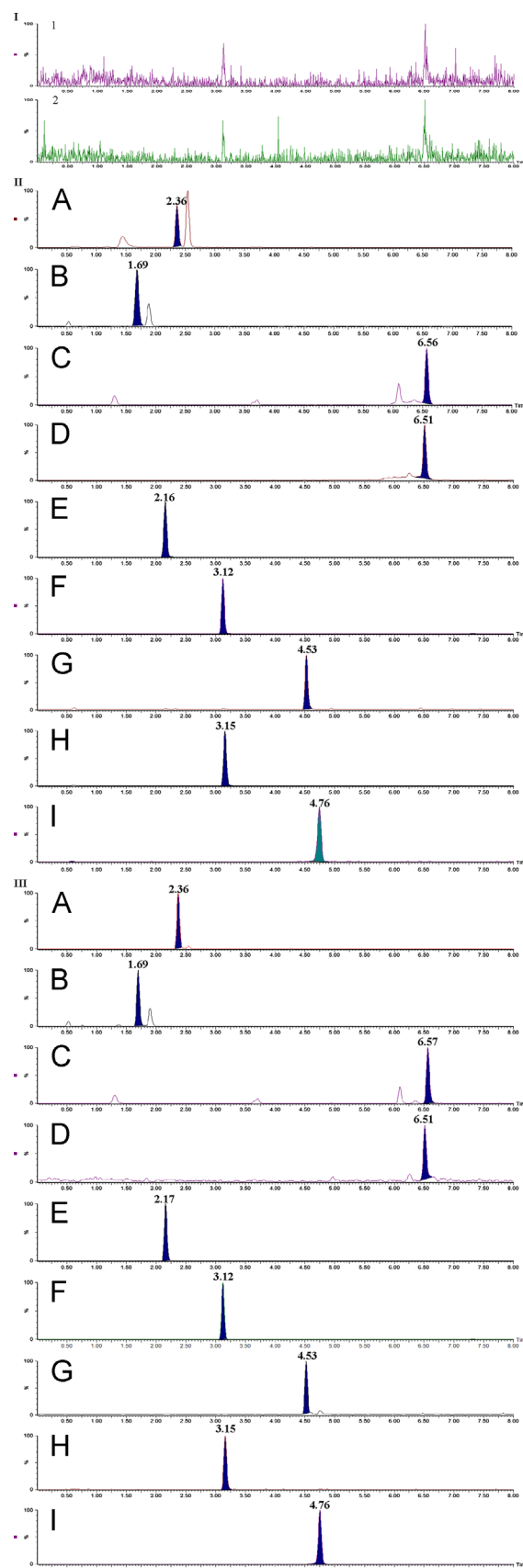
3.3. Precision and accuracy

The method showed good precision and accuracy with the intra-day and inter-day precision. All of results were found to be within the accepted variable limits and shown in Table 2.

Fig. 2. Representative MRM chromatograms of the four components in rats: I. Blank plasma sample (1. blank plasma sample from healthy rat; 2. blank plasma sample from blood deficiency rat). II. Blank plasma samples spiked with reference standards and internal standards. III. Rats plasma samples after oral administration of DBD. Note: (A) ferulic acid; (B) caffeic acid; (C) butylphthalide; (D) ligustilide; (E) calycosin-7-*O*- β -glucoside; (F) ononin; (G) astragaloside IV; (H) chloroamphenicol; and (I) clarithromycin.

3.4. Extraction recovery and matrix effect

Extraction recovery and matrix effect of the seven components were shown in Table 3. Average extraction recovery of clarithromycin and chloroamphenicol were $85.34 \pm 2.94\%$ and $87.69 \pm 4.39\%$, respec-



tively. Matrix effect of clarithromycin and chloroamphenicol were $89.78 \pm 4.31\%$ and $91.51 \pm 2.54\%$, respectively.

3.5. Stability

The stability of the analytes in rat plasma under different temperature and timing conditions was evaluated. The results showed that these analytes in rat plasma were all stable in different conditions. The results were shown in Table 4.

3.6. Pharmacokinetic study

3.6.1. The component contents in DBD

HPLC-TQ/MS was employed to detect the component contents in DBD. The results showed that DBD (1.0 g/mL) contained 2.806 $\mu\text{g/mL}$ of ferulic acid, 0.874 $\mu\text{g/mL}$ of caffeic acid, 0.083 $\mu\text{g/mL}$ of butylphthalide, 1.573 $\mu\text{g/mL}$ of ligustilide, 14.610 $\mu\text{g/mL}$ of calycosin-7-O- β -glucoside, 2.489 $\mu\text{g/mL}$ of ononin, and 3.275 $\mu\text{g/mL}$ of astragaloside IV.

3.6.2. Comparison of pharmacokinetic parameters of each DBD constituent in normal and blood deficiency rats

The mean plasma concentration–time profiles of the seven bioactive components were determined after oral administration of DBD extract, as were illustrated in Fig. 3, moreover, the partial pharmacokinetic parameters were shown in Table 5.

In Table 5, pharmacokinetic parameters of ferulic acid of blood deficiency rats after oral administration of DBD displayed higher C_{max} and $AUC_{0\sim T}$, and lower $T_{1/2Z}$ and $MRT_{0\sim T}$; and $T_{1/2Z}$ and $AUC_{0\sim T}$ had significant difference ($P < 0.05$). T_{max} of ferulic acid in normal and blood deficiency rats was same, which implied ferulic acid absorbed more rapidly in model rats. It indicated that blood deficiency model could reinforce bioavailability of ferulic acid. Primary study demonstrated that ferulic acid could nourish and tonify blood in our laboratory. Furthermore, ferulic acid was mainly absorbed in stomach and small intestine, poor blood circulation will prolong retention time in the stomach and small intestine and cause the increase of its absorption (Li et al., 2012a). As to caffeic acid, pharmacokinetic parameters of blood deficiency rats manifested higher C_{max} , $T_{1/2Z}$, and $MRT_{0\sim T}$, and lower T_{max} and $AUC_{0\sim T}$; and $T_{1/2Z}$ and $MRT_{0\sim T}$ had significantly different ($P < 0.05$). It indicated that blood deficiency model could slightly increase bioavailability of caffeic acid. Primary study in our laboratory also demonstrated that caffeic acid could nourish and tonify blood (Li et al., 2012b). Caffeic acid was mainly absorbed in intestine, and liver and blood were its main metabolic organs (Azuma et al., 2000; Lafay et al., 2006; Moridani et al., 2001). Therefore, it is impossible for caffeic acid to be metabolized quickly because of the pathological change in the liver and blood in model rats. For butylphthalide, pharmacokinetic parameters of blood deficiency rats revealed higher C_{max} , $T_{1/2Z}$, $AUC_{0\sim T}$ and $MRT_{0\sim T}$, and a lower T_{max} ; and only $MRT_{0\sim T}$ showed significant difference ($P < 0.05$). It might meant blood deficiency model could reinforce bioavailability of butylphthalide. Metabolic process of

Table 1

Calibration curve parameters and LLOQ of the DBD constituents in rat plasma samples.

Compounds	Regression equation	Correlation coefficient	linearity range (ng/mL)	LLOQ (ng/mL)
Ferulic acid	$Y=0.0032X+0.0324$	0.9980	8.85–354	8.150
Caffeic acid	$Y=0.0025X-0.0027$	0.9973	0.963–385	0.385
Butylphthalide	$Y=0.0083X-0.0023$	0.9999	1.31–262	1.090
Ligustilide	$Y=0.0009X+0.0233$	0.9917	0.73–730	0.690
Calycosin-7-O- β -glucoside	$Y=0.0747X+0.0017$	0.9968	0.728–291	0.676
Ononin	$Y=0.0461X+0.0276$	0.9937	1.69–338	0.845
Astragaloside IV	$Y=0.0002X+0.0001$	0.9994	15.95–319	12.87

Table 2

Intra-day, inter-day accuracy and precision of the DBD constituents in rat plasma samples.

Compounds	Spiked Concentration (ng/mL)	Intra-day			Inter-day		
		Measured (ng/mL)	RR %	RSD (%)	Measured (ng/mL)	RR %	RSD (%)
Ferulic acid	177	178.03 ± 10.17	100.58 ± 5.76	5.73	179.24 ± 9.92	101.27 ± 5.61	5.54
	35.4	34.89 ± 3.65	98.56 ± 10.33	10.48	35.82 ± 2.98	101.19 ± 8.42	8.32
	17.7	18.98 ± 1.74	107.23 ± 9.82	9.16	17.98 ± 0.99	101.58 ± 5.61	5.52
Caffeic acid	192.5	191.27 ± 11.39	99.36 ± 5.92	5.96	193.21 ± 8.88	100.37 ± 4.59	4.57
	38.5	36.99 ± 2.75	96.08 ± 7.15	7.44	39.1 ± 1.98	101.56 ± 5.15	5.07
	19.25	20.18 ± 1.02	104.83 ± 5.27	5.03	19.78 ± 1.73	102.75 ± 8.99	8.75
Butylphthalide	365	356.18 ± 21.84	97.58 ± 5.98	6.13	366.64 ± 20.16	100.45 ± 5.53	5.50
	73	69.75 ± 5.55	95.54 ± 7.62	7.98	73.67 ± 6.01	100.92 ± 8.24	8.16
	36.5	37.20 ± 2.34	101.92 ± 6.39	6.27	38.11 ± 2.35	104.41 ± 6.44	6.17
Ligustilide	131	130.09 ± 9.92	99.31 ± 7.53	7.58	132.19 ± 10.01	100.91 ± 7.62	7.55
	26.2	26.51 ± 2.76	101.18 ± 10.50	10.38	28.23 ± 3.12	107.75 ± 11.91	11.05
	13.1	12.89 ± 1.11	98.40 ± 8.45	8.59	14.10 ± 0.93	107.63 ± 7.11	6.61
Calycosin-7-O- β -glucoside	145.5	147.28 ± 10.58	101.22 ± 7.27	7.18	146.63 ± 9.73	100.78 ± 6.69	6.64
	29.1	30.10 ± 3.09	103.44 ± 10.60	10.24	32.7 ± 2.56	112.37 ± 8.80	7.83
	14.55	15.69 ± 1.25	107.84 ± 8.59	7.97	14.35 ± 1.44	98.63 ± 9.90	10.04
Ononin	169	170.77 ± 11.62	101.05 ± 6.88	6.81	168.89 ± 8.73	99.93 ± 5.17	5.17
	33.8	35.27 ± 4.06	104.35 ± 12.03	11.53	33.89 ± 3.87	100.27 ± 11.46	11.43
	16.9	16.94 ± 1.33	100.24 ± 7.84	7.82	17.23 ± 1.75	100.24 ± 10.36	10.34
Astragaloside IV	159.5	162.01 ± 9.98	101.57 ± 6.27	6.17	160.00 ± 10.02	100.31 ± 6.27	6.25
	31.9	29.99 ± 3.01	94.01 ± 9.44	10.04	31.80 ± 3.06	99.69 ± 9.57	9.60
	15.95	15.77 ± 1.16	98.87 ± 7.29	7.37	15.98 ± 0.92	100.19 ± 5.77	5.76

Table 3

Extract recovery and matrix effect of the DBD constituents in rat plasma samples.

Compounds	Spiked concentration (ng/mL)	Extraction recovery (%)	RSD (%)	Matrix effect (%)	RSD (%)
Ferulic acid	177	81.74 ± 4.21	5.15	85.14 ± 3.77	4.43
	35.4	84.21 ± 5.33	6.33	82.98 ± 6.87	8.28
	17.7	69.66 ± 3.33	4.78	76.02 ± 2.95	3.88
Caffeic acid	192.5	74.43 ± 5.31	7.13	72.14 ± 5.13	7.11
	38.5	79.35 ± 3.84	4.84	88.78 ± 4.14	4.66
	19.25	62.96 ± 4.03	6.40	74.56 ± 3.22	4.32
Butyl phthalide	365	69.56 ± 4.05	5.81	72.99 ± 5.61	7.69
	73	67.85 ± 5.26	7.75	70.82 ± 4.99	7.05
	36.5	61.28 ± 2.63	4.29	63.24 ± 3.16	5.00
Ligustilide	131	68.96 ± 2.28	3.31	69.17 ± 2.17	3.14
	26.2	62.12 ± 4.83	7.78	66.38 ± 3.14	4.73
	13.1	56.82 ± 2.48	4.36	59.99 ± 5.65	9.42
Calycosin-7-O-β-glucoside	145.5	84.31 ± 2.61	3.10	91.21 ± 5.14	5.64
	29.1	78.61 ± 5.13	6.53	84.16 ± 4.98	5.92
	14.55	69.77 ± 4.54	6.51	87.19 ± 6.31	7.24
Ononin	169	79.56 ± 6.62	8.32	78.16 ± 4.69	6.00
	33.8	74.26 ± 5.82	7.84	81.21 ± 6.23	7.67
	16.9	65.50 ± 2.91	4.44	77.23 ± 2.22	2.87
Astragaloside IV	159.5	73.52 ± 3.93	5.35	73.78 ± 4.03	5.46
	31.9	69.52 ± 3.17	4.56	77.13 ± 4.66	6.04
	15.95	65.59 ± 3.48	5.31	64.14 ± 5.78	9.01

Table 4

Stability of the DBD constituents in rat plasma samples.

Compounds	Spiked Concentration (ng/mL)	Short-term		Freeze-thaw		Long-term	
		Measured (ng/mL)	RR (%)	Measured (ng/mL)	RR (%)	Measured (ng/mL)	RR (%)
Ferulic acid	177	178.22 ± 1.28	100.69	176.20 ± 2.11	99.55	177.14 ± 1.39	100.07
	35.4	35.20 ± 0.20	99.44	34.90 ± 0.19	98.59	35.46 ± 0.79	100.17
	17.7	17.78 ± 0.04	100.45	16.90 ± 0.15	95.48	18.23 ± 1.01	102.99
Caffeic acid	192.5	193.15 ± 0.97	100.33	192.79 ± 0.78	100.15	192.37 ± 0.99	99.93
	38.5	38.45 ± 0.88	99.87	39.21 ± 0.11	101.84	38.20 ± 0.29	99.22
	19.25	19.22 ± 0.17	99.84	19.28 ± 0.09	100.16	19.79 ± 0.04	102.81
Butylphthalide	365	367.29 ± 2.02	100.63	368.47 ± 1.09	100.95	366.22 ± 1.00	100.33
	73	73.28 ± 0.89	100.38	74.12 ± 0.98	101.53	74.01 ± 0.49	101.38
	36.5	36.40 ± 0.29	99.73	35.73 ± 0.74	97.89	36.76 ± 1.24	100.71
Ligustilide	131	131.20 ± 0.78	100.15	132.10 ± 1.19	100.84	132.38 ± 0.94	101.05
	26.2	26.78 ± 0.26	102.21	25.99 ± 0.76	99.20	26.67 ± 0.84	101.79
	13.1	13.99 ± 0.01	106.79	12.98 ± 0.22	99.08	13.83 ± 1.39	105.57
Calycosin-7-O-β-glucoside	145.5	145.20 ± 0.08	99.79	145.53 ± 0.21	100.02	146.97 ± 0.30	101.01
	29.1	29.73 ± 0.19	102.16	29.75 ± 0.24	102.23	29.98 ± 0.29	103.02
	14.55	15.20 ± 0.01	104.47	14.98 ± 0.08	102.96	14.83 ± 0.22	101.92
Ononin	169	170.10 ± 0.89	100.65	171.12 ± 0.54	101.25	168.34 ± 1.93	99.61
	33.8	35.01 ± 0.29	103.58	32.15 ± 0.18	95.12	34.83 ± 1.02	103.05
	16.9	16.84 ± 0.06	99.64	17.11 ± 0.29	101.24	16.29 ± 0.05	96.39
Astragaloside IV	159.5	161.58 ± 1.12	101.30	159.97 ± 0.19	100.29	158.38 ± 0.29	99.30
	31.9	32.19 ± 0.77	100.91	31.75 ± 0.98	99.53	30.99 ± 0.08	97.15
	15.95	16.98 ± 0.65	106.46	16.24 ± 1.01	101.82	16.07 ± 0.04	100.75

butylphthalide mainly happened in liver microsomes (Peng and Zhou, 1996), which referred that blood deficiency model could slow down the eliminated time of butylphthalide because of the pathological change in the liver. As to ligustilide and ononin, pharmacokinetic parameters of blood deficiency rats showed a higher C_{max} , and lower T_{max} , $T_{1/2}$, $AUC_{0\sim T}$ and $MRT_{0\sim T}$. It suggested that the blood deficiency model could speed up the time and extent of absorption and inhibit elimination of ligustilide and ononin, but the extent of absorption was lower, which might attributed to the biological changes of some enzymes and the biomembrane transfer abilities under pathological conditions (Kim et al., 2006; Tian et al., 2010). Studies showed that butylphthalides of DBD had the effects of dilating blood vessels and antiplatelet aggregation (Li et al., 2009; Kobayashi et al., 1993). In addition, shorter $T_{1/2}$ and $MRT_{0\sim T}$ implied the slower elimination of calycosin-7-O-β-glucoside in blood deficiency rats. Similarly to butylphthalide, astragaloside IV was also absorbed very fast and eliminated slow. Based on our knowledge, the present study about

pharmacokinetic of ononin and calycosin-7-O-β-glucoside had not been reported, no matter monomeric compounds, in TCM or in TCM prescriptions.

3.6.3. The pharmacokinetic comparison of different types of DBD constituents in normal and blood deficiency rats

Ingredients in TCM were complicated and abundant, and the pharmacokinetics of single ingredient was not able to represent the pharmacokinetics of the whole herbal medicine. For the sake of more representation, the total T_{max} and $AUC_{0\sim T}$ calculated from the three different types of ingredients in DBD as two new parameters were used to compare their pharmacokinetic profiles of normal and blood deficiency rats. The total T_{max} and $AUC_{0\sim T}$ of aromatic acids (ferulic acid and caffeic acid) were 70 min, 8930.6 ng min/mL in normal rats and 55 min, 9878.77 ng min/mL in blood deficiency rats, respectively. The total T_{max} and $AUC_{0\sim T}$ of phthalides (butylphthalide and ligustilide) were 23.5 min, 9289.83 ng min/mL in normal rats and 20 min,

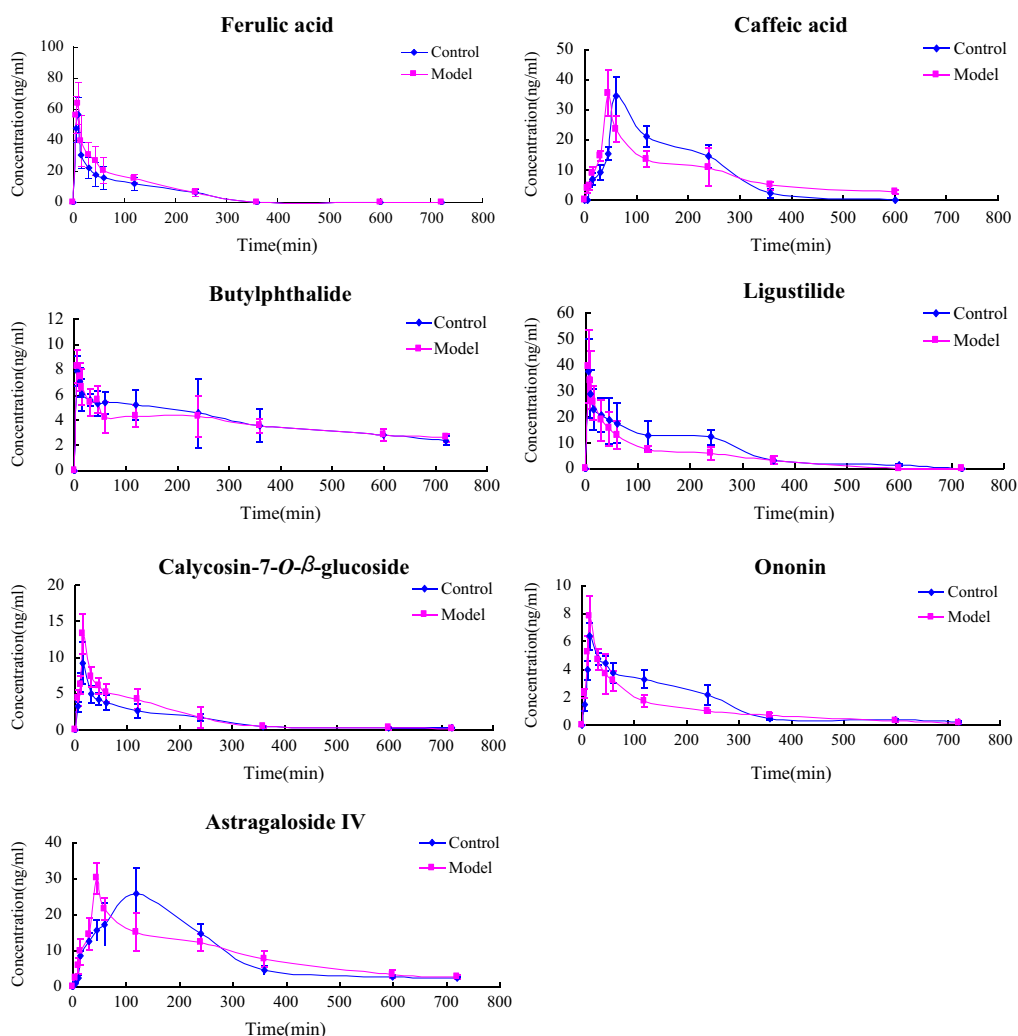


Fig. 3. Mean plasma concentration–time curves of seven components after oral administration of DBD to normal and blood deficiency rats ($n=6$).

Table 5

NCA pharmacokinetic parameters of seven compounds of DBD between normal and blood deficiency rats (mean $\pm$ SD, $n=6$).

Compounds	Group	$C_{\max}$ (ng/mL)	$T_{\max}$ (min)	$T_{1/2}$ (min)	$AUC_{0\sim T}$ (ng min/mL)	$MRT_{0\sim T}$ (min)
Ferulic acid	Control	61.03 $\pm$ 8.04	10.00 $\pm$ 0.00	277.61 $\pm$ 8.79	3341.44 $\pm$ 688.80	83.89 $\pm$ 8.65
	Model	68.90 $\pm$ 9.61	10.00 $\pm$ 0.00	99.42 $\pm$ 33.17**	4412.54 $\pm$ 271.36*	81.46 $\pm$ 0.53
Caffeic acid	Control	34.43 $\pm$ 6.25	60.00 $\pm$ 0.00	77.24 $\pm$ 24.66	5589.16 $\pm$ 634.56	142.51 $\pm$ 13.10
	Model	35.52 $\pm$ 7.39	45.00 $\pm$ 0.00	194.68 $\pm$ 119.04	5466.23 $\pm$ 850.10	191.32 $\pm$ 14.02**
Butylphthalide	Control	7.35 $\pm$ 0.92	14.75 $\pm$ 6.60	809.73 $\pm$ 224.89	4196.69 $\pm$ 179.69	528.48 $\pm$ 16.08
	Model	7.93 $\pm$ 1.03	12.50 $\pm$ 2.89	1421.22 $\pm$ 493.57	4397.68 $\pm$ 473.00	601.51 $\pm$ 46.53*
Ligustilide	Control	34.18 $\pm$ 6.81	8.75 $\pm$ 4.79	168.88 $\pm$ 39.01	5093.14 $\pm$ 1044.13	174.24 $\pm$ 8.37
	Model	35.06 $\pm$ 10.90	7.50 $\pm$ 2.89	140.51 $\pm$ 52.53	3063.00 $\pm$ 453.82*	119.87 $\pm$ 16.25**
Calycosin-7-O- β -glucoside	Control	9.28 $\pm$ 2.87	15.00 $\pm$ 0.00	516.47 $\pm$ 72.62	1179.10 $\pm$ 93.84	279.68 $\pm$ 36.66
	Model	14.58 $\pm$ 1.13*	15.00 $\pm$ 0.00	260.98 $\pm$ 153.14*	1442.20 $\pm$ 278.21	223.53 $\pm$ 19.69*
Ononin	Control	6.42 $\pm$ 0.90	22.50 $\pm$ 15.00	423.80 $\pm$ 97.16	1229.19 $\pm$ 124.55	263.69 $\pm$ 14.45
	Model	7.83 $\pm$ 1.49	15.00 $\pm$ 0.00	170.72 $\pm$ 20.56*	827.60 $\pm$ 97.21**	176.94 $\pm$ 7.38**
Astragaloside IV	Control	24.70 $\pm$ 1.73	105.00 $\pm$ 30.00	897.80 $\pm$ 915.06	8365.48 $\pm$ 818.29	383.60 $\pm$ 20.03
	Model	33.99 $\pm$ 0.06*	45.00 $\pm$ 0.00*	630.04 $\pm$ 378.24	8576.21 $\pm$ 402.77	422.09 $\pm$ 19.37*

* $P < 0.05$

** $P < 0.01$, compared with normal group.

7460.68 ng min/mL in blood deficiency rats, respectively. And the total $T_{\max}$ and $AUC_{0\sim T}$ of flavonoid glycosides (ononin, calycosin-7-O- β -glucoside and astragaloside IV) were 142.5 min, 10773.77 ng min/mL in normal rats and 75 min, 10846.01 ng min/mL in blood deficiency rats, respectively. From the two new parameters, we could know that $T_{\max}$ of three different types of ingredients in blood

deficiency rats were all shorter than normal rats. And $AUC_{0\sim T}$ of aromatic acids and flavonoid glycosides in blood deficiency rats were larger than normal rats, but that of phthalides were smaller. The main explanation for these differences seems to be the variances of component content in blood samples, which might be caused by the differences of metabolic capacity in normal and blood deficiency.

These data indicated that blood deficiency model could reinforce bioavailability of some aromatic acids and flavonoid glycosides in DBD. Usually medicine could have onset in pathological state, so the *in vivo* process in normal and pathological state might be the causes of their different biological activities. Compared the gender differences in absorption of normal group and blood deficiency group, the results showed the pharmacokinetic parameters of male and female in normal and blood deficiency rats were different, but there was no significant difference between them.

Diagnosis and treatment based on an overall analysis of the illness and the patient's condition was the essence of TCM theory. Rational drug therapy was dependent upon a basic understanding of the way patients handle drugs (pharmacokinetics) and their response to specific drug effects (pharmacodynamics). Moreover, traditional Chinese syndrome state in traditional Chinese recipe could significantly influence the blood drug concentration and their pharmacokinetic parameters after oral administration (Huang et al., 2000; Tian et al., 2010). Usually medicine could have onset after entering the body, so the *in vivo* process of medicines might be the most possible causes of the different efficacies (Feng et al., 2010). More and more researches were focused on the pharmacokinetics of ingredients in TCM in recent years, and the main strategy was to choose one or several effective ingredients as the target components. This study was aimed at comparing the difference of the same ingredients between different animals. Furthermore, blood deficiency model in many studies was conducted by radiation and chemical injury based on our knowledge (Li et al., 2012b; Huo et al., 2010), this study was the first report about pharmacokinetic investigation in blood deficiency animals which was conducted by bleeding. Results of the present study indicated the potential pharmacokinetic difference between normal and blood deficiency rats.

Different from modern disease, TCM symptom concentrates on the overall state in patient, which was becoming a trend in modern medicine for treating complicated diseases. Therefore, further studies about different TCM symptoms were urgent in spite of the existent difficulties of model-induced. Many pharmacokinetic studies focused on the normal animals, it might be unreasonable for the research of TCM metabolism. As we all knew, every herb had many ingredients, and different ingredient had their own pharmacokinetic process. So, it was important to take all of pharmacokinetic processes in normal and pathological states into consideration to achieve the maximize drug effects.

4. Conclusion

In the present work, a UPLC-TQ/MS method for simultaneous determination of plasma concentrations of seven bio-active constituents of DBD was developed and validated. The results showed that the method was selective, sensitive, precise, accurate, and reliable for comparative quantification of ferulic acid, caffeic acid, butylphthalide, ligustilide, calycosin-7-O- β -glucoside, ononin and astragaloside IV in normal and blood deficiency rat plasma samples. The results demonstrated that the seven DBD constituents in normal and blood deficiency rats had obvious differences in some pharmacokinetic characteristics, suggesting that the rate and extent of drug metabolism was altered in blood deficiency animals. The study provided many valuable data for illuminating the variation mechanism of nourishing and tonifying blood after oral administration of DBD, and for better clinical applications of DBD and the relative TCM formulae.

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